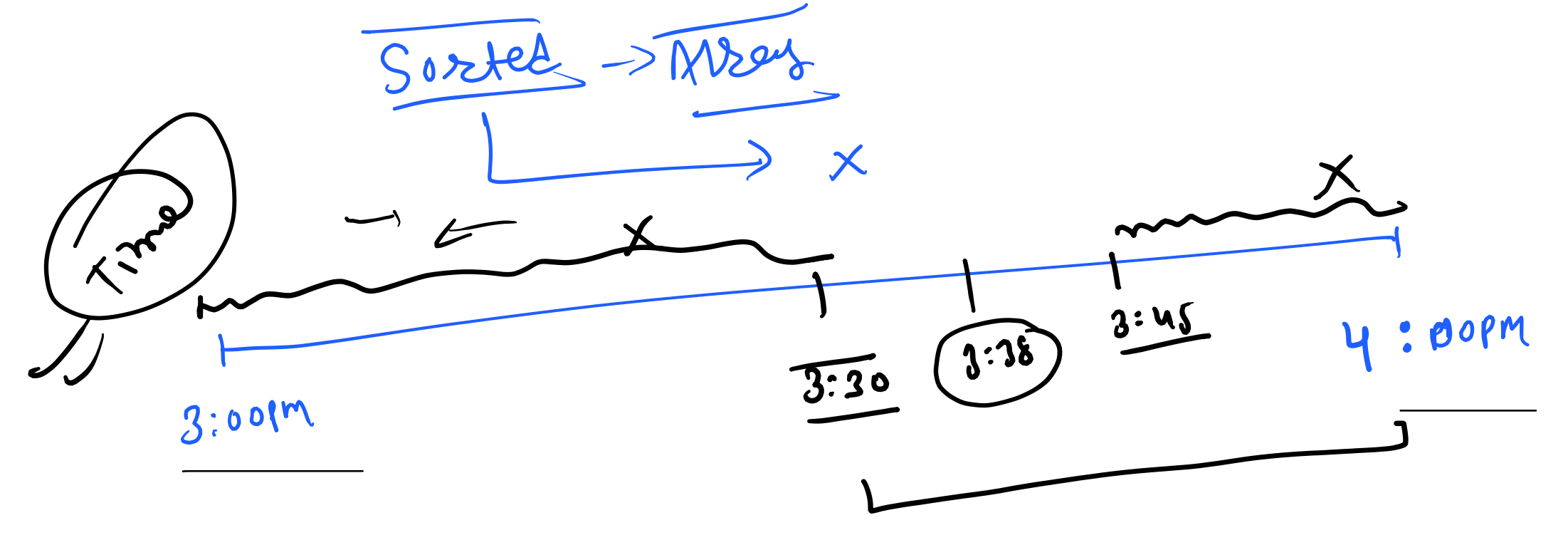
Lec-10

26/09/26 4:44 PM

Linear search / Binary search

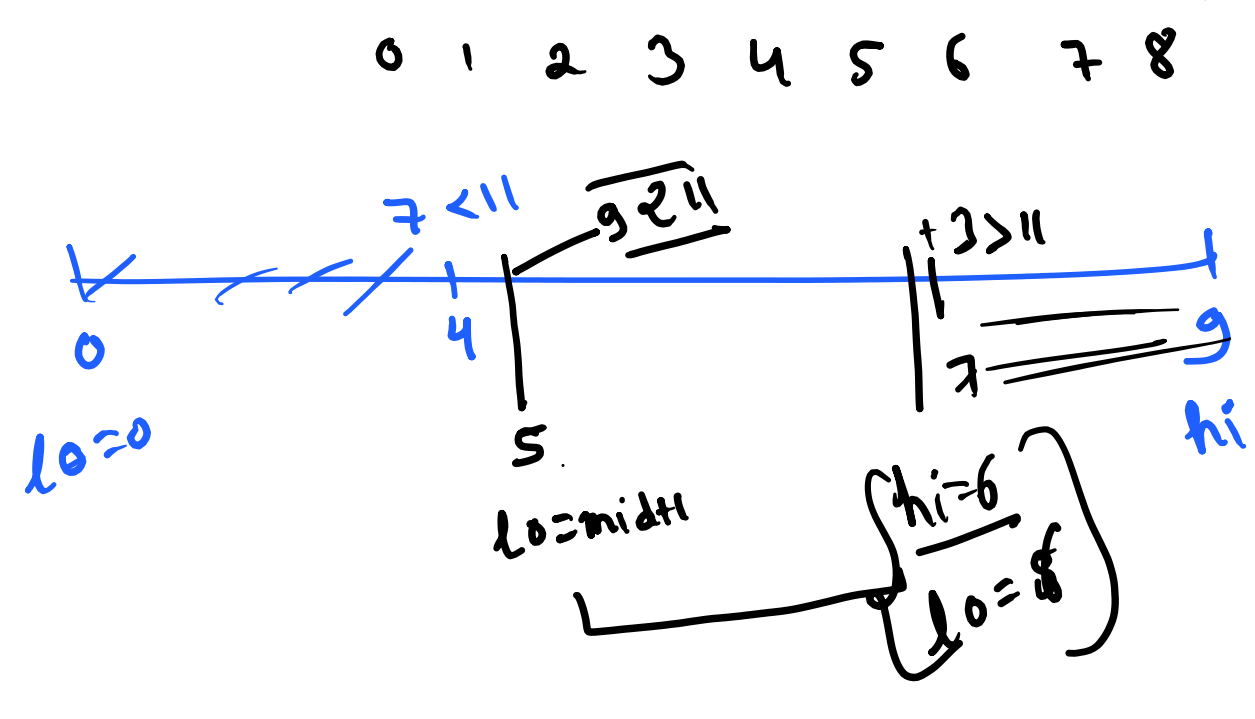
Sorted -> Array


Time
3:00PM
3:30
3:38
3:45
4:00PM

Algo

item = 11

[2, 3, 4, 5, 7, 9, 11, 13, 18, 19]
0 1 2 3 4 5 6 7 8 9


7<11
9<11
13>11
0
4
5
7
9
lo=0
lo=mid+1
hi=6
lo=6
hi

```
lo=0, hi=arr.len-1
While (lo <= hi) {
    mid = (lo+hi)/2
    if (arr[mid]==item) {
        return mid
    }
    else if (arr[mid] > item) {
        hi = mid-1
    }
    else {
        lo = mid+1
    }
}
```

maximum ← $x^K <= N$

$1^3 \le 149$
$2^3 \le 149$
$3^3 \le 149$
$4^3 \le 149$
$5^3 \le 149$
$6^3 \le 149$ X

x=5

K=3
N=149

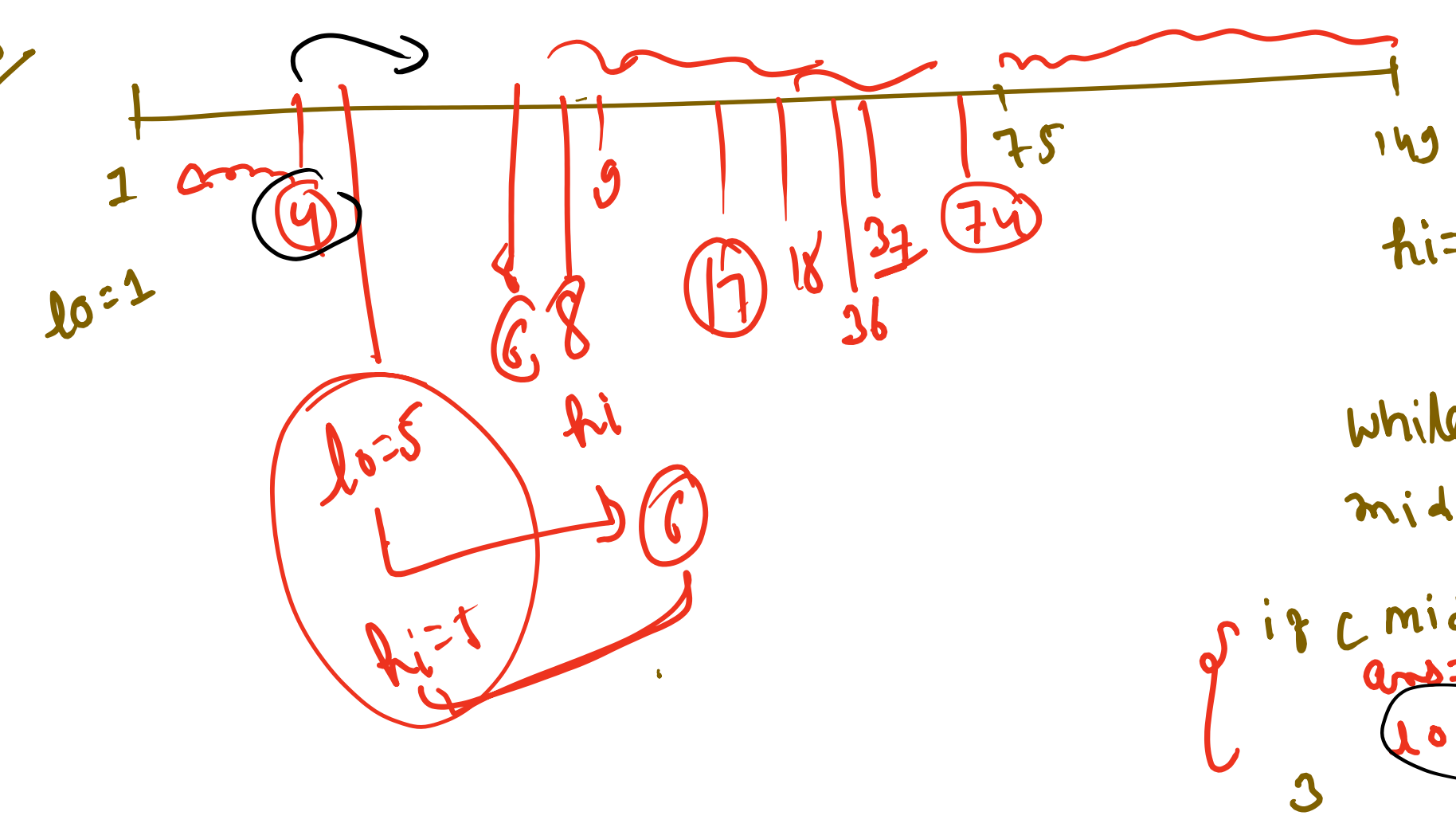
```
x=1
while (x^K <= n) {
    x++
}
return x-1
```

$x^K \le N$

N=149 ans=~~4~~ 5

$4^3 \le 149$

~~$75^3 \le 149$~~
~~$37^3 \le 149$~~

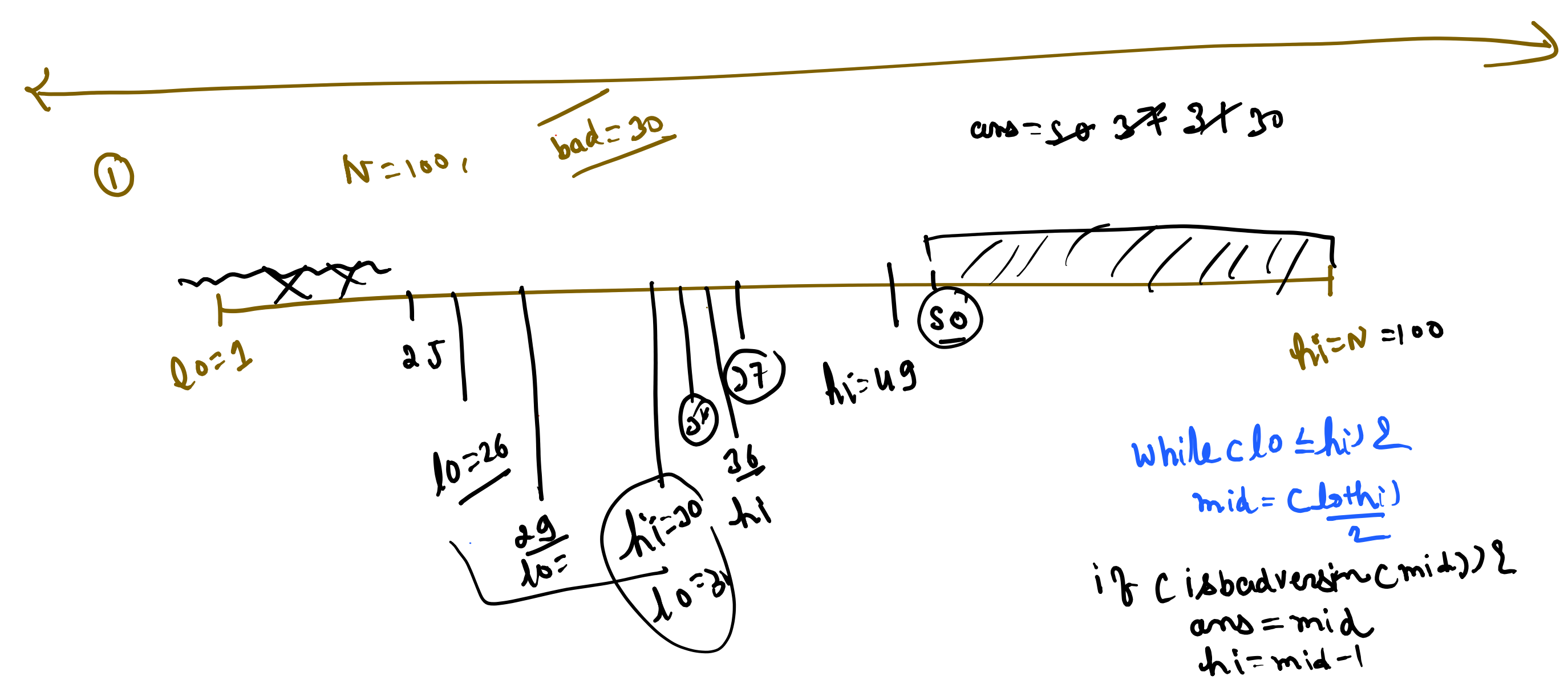

1
4
lo=1
6
8
9
17
18
37
36
74
75
149
hi
hi=N
lo=5
hi=7
6

ans=5

```
while (lo <= hi) {
    mid = (lo+hi)/2
    if (mid^K <= n) {
        ans = mid
        lo = mid+1
    }
    else {
        hi = mid-1
    }
}
```

① N=100, bad=30

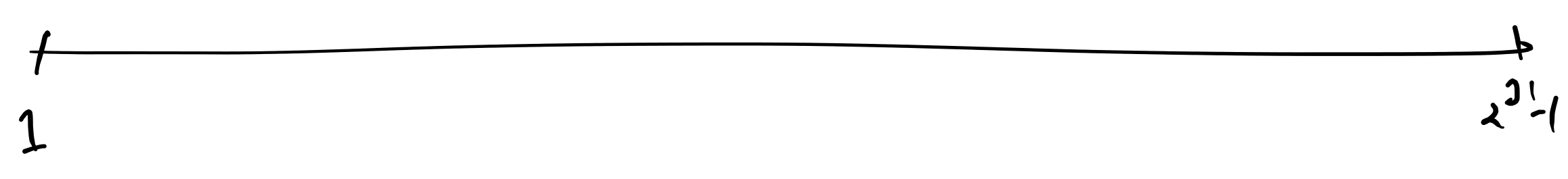
ans = ~~50~~ ~~37~~ ~~31~~ 30


lo=1
25
lo=26
29
lo=
hi=30
lo=31
34
37
36
hi
hi=49
50
hi=N=100

```
while (lo <= hi) {
    mid = (lo+hi)/2
    if (isbadversion(mid)) {
        ans = mid
        hi = mid-1
    }
    else {
        lo = mid+1
    }
}
```

$N = 2^{31}-1$ $bad = 2^{31}-3$

$mid = lo + \frac{(hi-lo)}{2} = \frac{2lo+hi-lo}{2}$



$lo = 2^{31}-6$ $hi = 2^{31}-1$

$mid = \frac{(lo+hi)}{2} = \frac{2^{31}-6+2^{31}-1}{2} = \frac{2\times2^{31}-7}{2}$ number overflow

$2^{31}-1$

$mid = \frac{lo}{2} + \frac{hi}{2}$, $\frac{7}{2} + \frac{11}{2}$ → 3 + 5 = 8

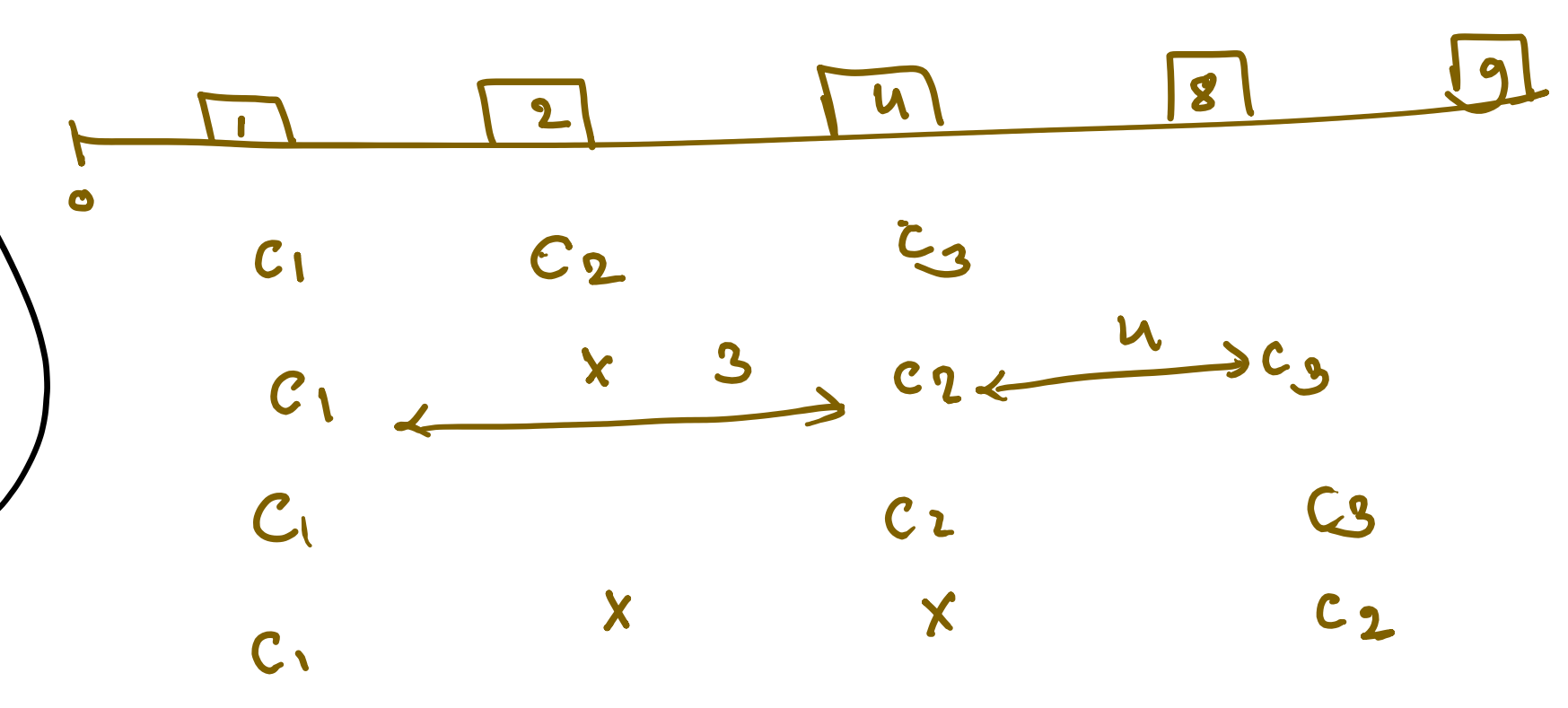
[1, 2, 4, 8, 9]

Nos → noc

```
5 3
1
2
8
4
9
```

3

min=1
min=2
min=3
min=4


0
1
2
4
8
9
C1 C2 C3
C1 x 3 C2 4 C3
C1 C2 C3
C1 x x C2

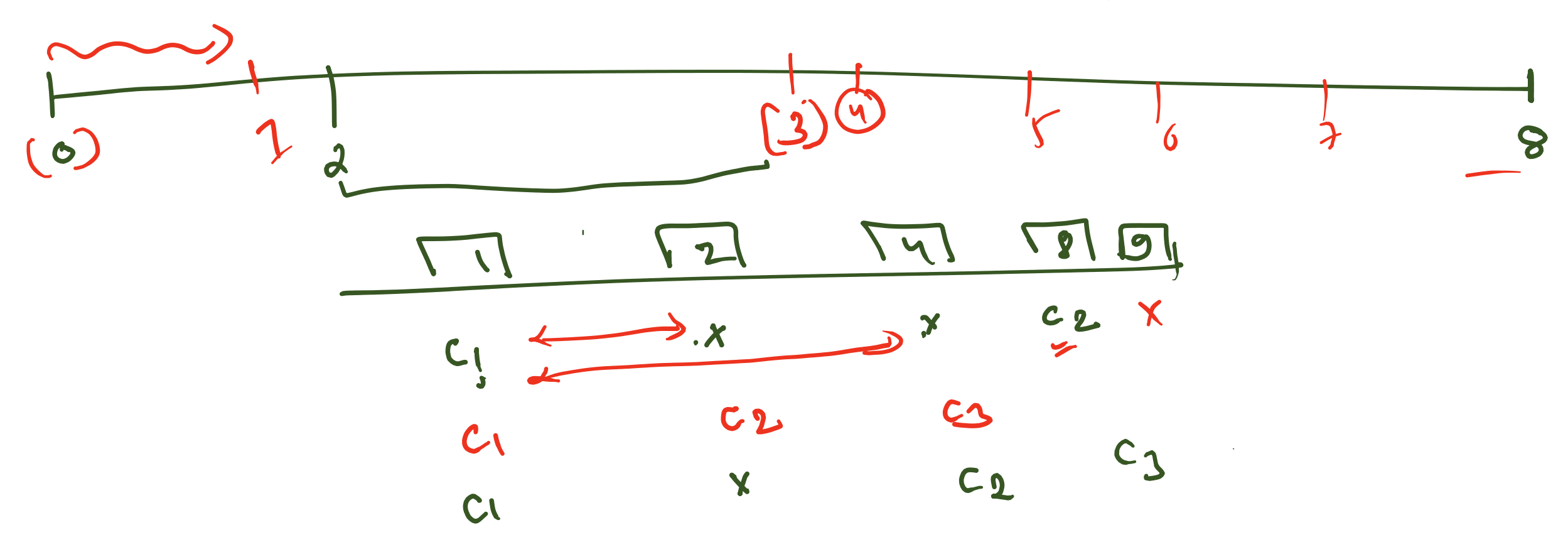
ans = ~~1~~ 2 3

ans = ~~1~~ 2

[1, 2, 4, 8, 9]

```
min=1
while (isitpossible(min)) {
    min++
}
```

(10, 20, 40, 80, 90)

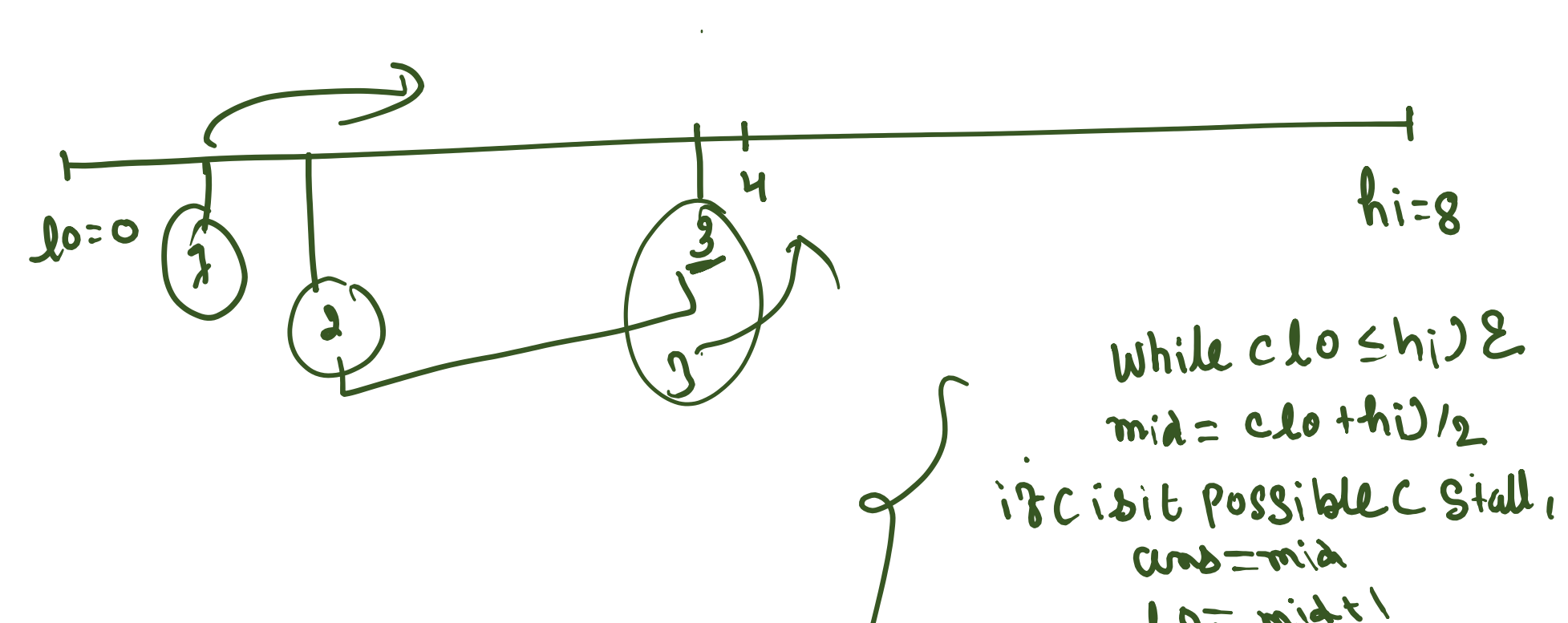

(0)
1
2
3
4
5
6
7
8
1
2
4
8
9
C1 x x C2 x
C1 C2 C3
C1 x C2 C3

[1, 2, 4, 8, 9]
0 1 2 3 4

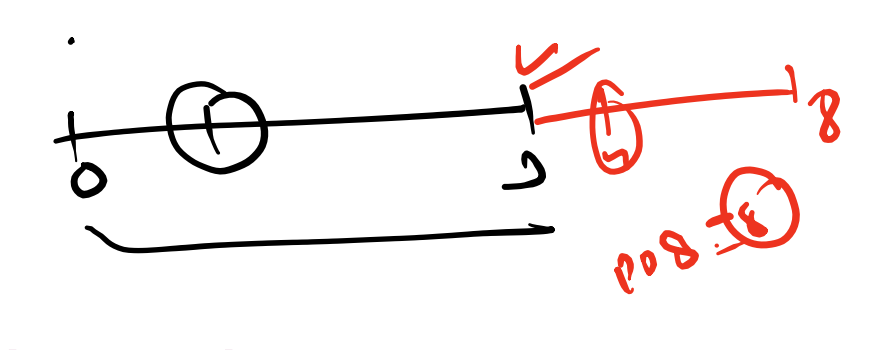
lo=0

hi = stall[N-1] - stall[0]

ans=2


lo=0
1
2
3
4
hi=8

```
while (lo <= hi) {
    mid = (lo+hi)/2
    if (isitpossible(stall, noc, mid)) {
        ans = mid
        lo = mid+1
    }
    else {
        hi = mid-1
    }
}
```



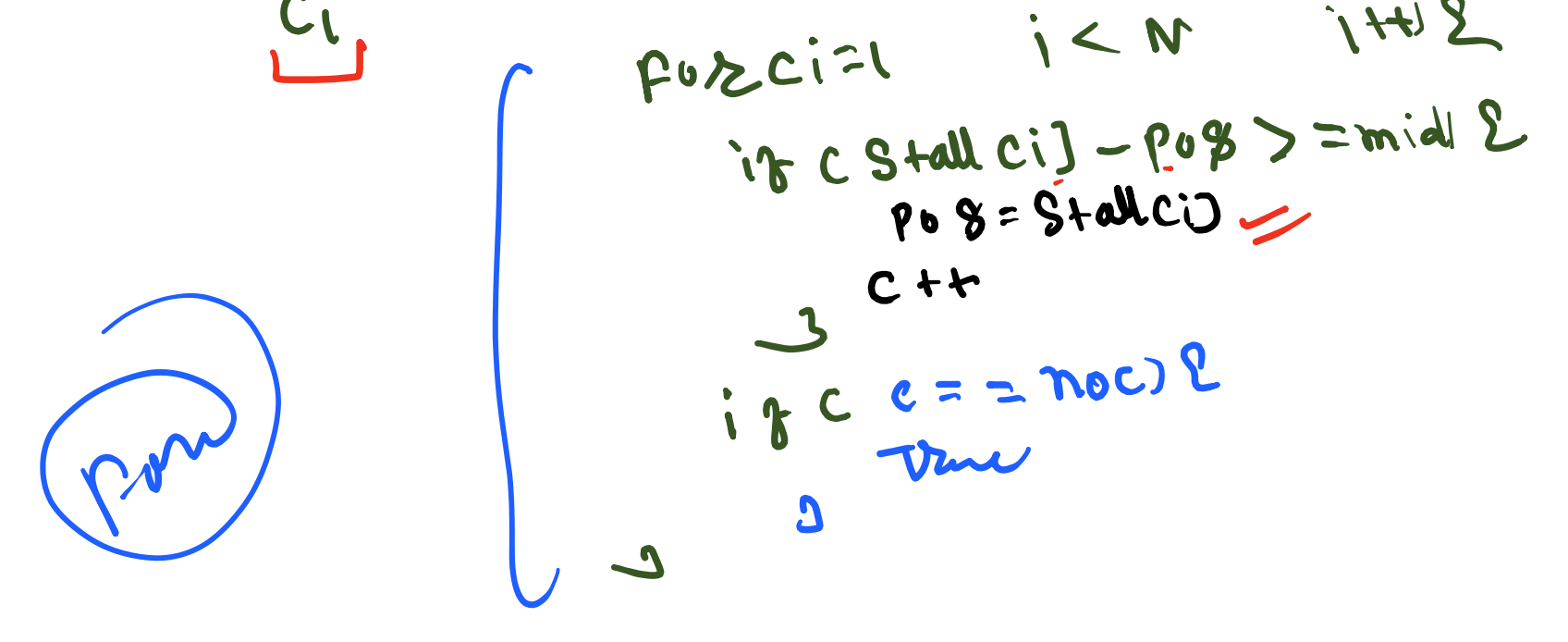
[1, 2, 4, 8, 9]
0 1 2 3 4

2-1>=1
4-2>=1

```
public static boolean isitpossible(int[] stall, int noc, int mid) {
    // TODO Auto-generated method stub

}
```

pos = stall[0] 4
c = ~~1~~ 2 3
mini = 4


4
1
2
4
8
9
C1
C2

```
for (i=1; i<N; i++) {
    if (stall[i] - pos >= mid) {
        pos = stall[i]
        c++
    }
    if (c == noc) {
        True
    }
}
```

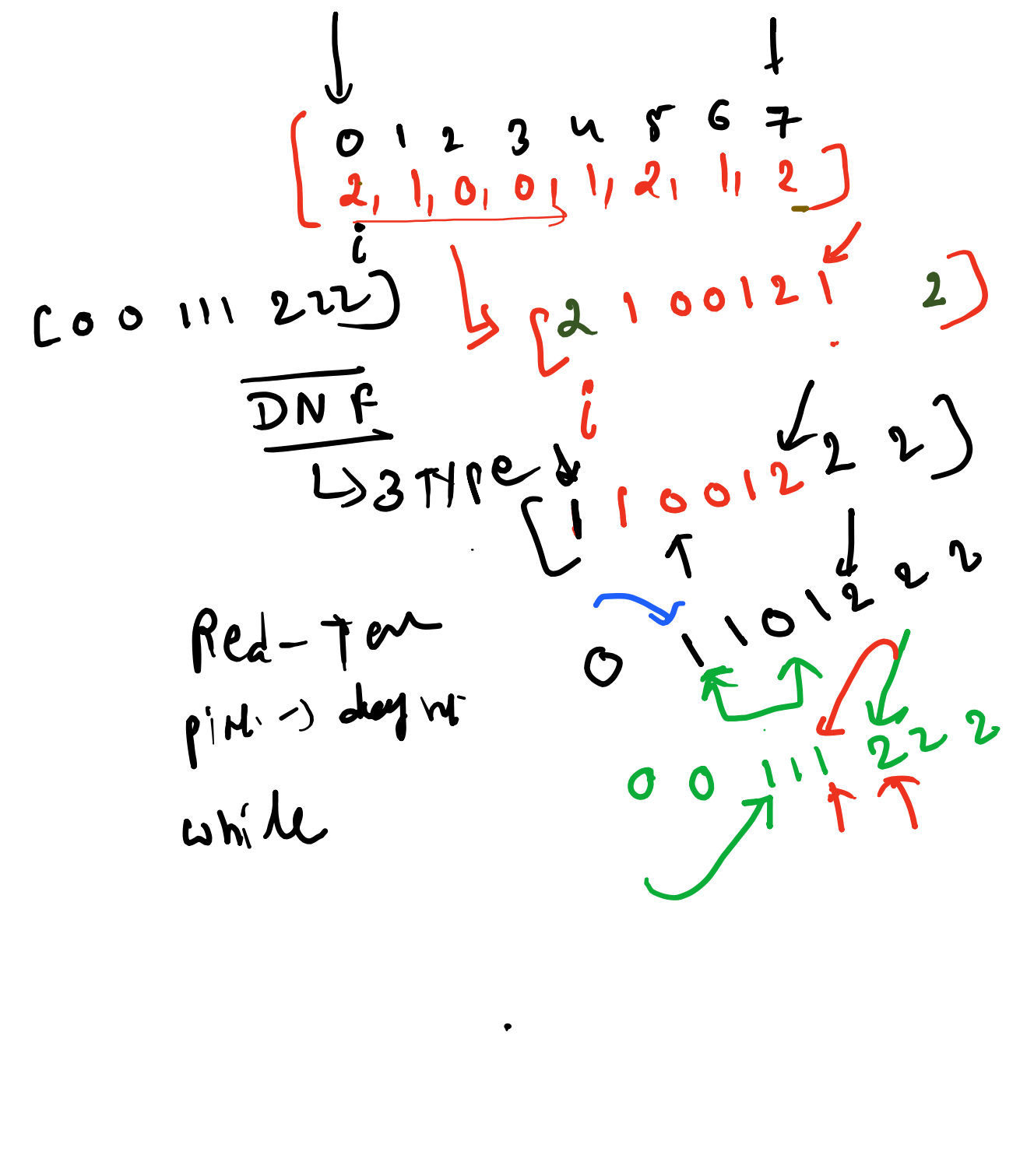
pow

[0 0 111 222]

DNF
↳ 3 type

Red-Tan
pin -> day w
while

2->3
1->3
0->2


zero
Two
0 1 2 3 4 5 6 7
2, 1, 0, 0, 1, 2, 1, 2
i
[2 1 0 0 1 2 1 2]
[1 1 0 0 1 2 2 2]
0 1 1 0 1 2 2 2
0 0 1 1 1 2 2 2

Two = ~~7~~ ~~6~~ 5

```
while (i <= Two) {
    if (arr[i]==0) {
        swap(i, zero)
        zero++
        i++
    }
    else if (arr[i]==2) {
        swap(i, two)
        Two--
    }
    else {
        i++
    }
}
```

```
public static void Sort(int[] arr) {
    int zero = 0;
    int two = arr.length - 1;
    int i = 0;
    while (i <= two) {
        if (arr[i] == 0) {
            int temp = arr[zero];
            arr[zero] = arr[i];
            arr[i] = temp;
            zero++;
            i++;
        } else if (arr[i] == 2) {
            int temp = arr[two];
            arr[two] = arr[i];
            arr[i] = temp;
            two--;

        } else {
            i++;
        }
    }
}
```

[0, 1, 1, 2, 2, 0]
[0 1 1 2 2 0]
i
[0, 1, 1 0 2 2]